Selva Ganesh V

Full Stack Engineer | React.js · TypeScript · NestJS · PostgreSQL
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Professional Summary

Full Stack Software Engineer with 3+ years of experience building enterprise platforms using React.js, TypeScript, Node.js, NestJS, and React Native. Experienced in developing scalable frontend applications, designing RESTful APIs, implementing workflow engines, graph-processing systems, and real-time features using Socket.IO.

Technical Skills

Languages: JavaScript (ES6+), TypeScript
Frontend: React.js, Next.js, React Native, Expo, Electron, Zustand, TanStack Query, TanStack Router, TanStack Start, Svelte, React Flow, HTML5, CSS3, Vite, TailwindCSS
Backend: Node.js, NestJS, Express.js, Hono, ElysiaJS, RESTful APIs, WebSockets
Databases: PostgreSQL, MongoDB, Redis, TypeORM, Drizzle ORM
Authentication & Security: JWT, RBAC, BetterAuth
Tools & Infrastructure: Git, GitHub, TurboRepo (Monorepo), Socket.IO, Swagger/OpenAPI, Jest, Agile/Scrum, CI/CD
Architecture & Concepts: Full Stack Development, Real-Time Collaboration, Workflow Engines, Schema-Driven Systems, Low-Code Platforms, Graph Processing, System Design

Professional Experience

Software Development Engineer Aug 2026 – Present
AGG Technology Private Limited *Chennai, TN, India*

- Developed a multi-sport tournament platform across **four TypeScript applications** — NestJS backend, two Next.js web apps, and an Expo/React Native mobile app — covering live scoring, registrations, auctions, and leaderboards.
- Hardened authentication with global JWT, tournament-role, and rate-limiting guards across **35 NestJS controllers**, adding single-use refresh-token rotation and cross-tab token refresh via Web Locks and BroadcastChannel.
- Architected an external developer API of **31 REST endpoints** with SHA-256-hashed keys, scoped permissions, monthly quotas, and usage metering — validated by 57 Jest tests and OpenAPI documentation.
- Eliminated concurrent score-write races across four sport engines using TypeORM transactions with **11 pessimistic row locks**, replacing 1-second polling with per-match timers over a singleton Socket.IO gateway.
- Delivered tournament registration end-to-end — NestJS cart service, Next.js wizard with DUPR OAuth autofill and PDF receipts, and React Native checkout — backed by a **391-line PostgreSQL recursive CTE** deduplicating players across three identity schemes.
- Engineered an offline-capable React Native data layer with TanStack Query persistence and a typed axios client over **41 REST endpoints**, with Socket.IO-driven live-score cache invalidation across a 48-route app.
- Built a real-time cricket and pickleball auction platform with a three.js WebGL arena, public LED stage, and organizer console wired to **9 WebSocket events** and an 8-second bid watchdog.
- Shipped tournament email broadcasting with recipient dedupe, HTML sanitization, and chunked SMTP delivery, alongside a config-driven leaderboard with 18 ranking criteria.

Software Engineer Oct 2023 – Apr 2026
Torus Innovations *Tirunelveli, TN, India*

- Developed a large-scale enterprise low-code platform using **React.js, TypeScript, NestJS, PostgreSQL, and Socket.IO** for fintech clients across the UAE and Kenya.
- Built a visual flow builder with **13+ diagram types** (DFD, ERD, PFD, cloud deployment, API flow) and **40+ custom ReactFlow nodes**, with snapshot-based undo/redo (100-state history) and custom edge routing.
- Engineered a WYSIWYG UI editor with DND Kit (custom collision detection, parent-child reparenting), alignment guides, and snapping — delivering **20+ production banking screens** backed by a **50+ component library**.
- Designed **35+ RESTful APIs and NestJS microservice message patterns** across a multi-tenant hierarchy, covering workflow orchestration, artifact management, and field-level encryption with RBAC.
- Implemented workflow validation engines using graph-processing to prevent circular dependencies; built artifact locking and cross-tab presence sync via Socket.IO and BroadcastChannel.

Projects

Vezal.db | *React, TypeScript, TanStack Router, Hono, Drizzle, PostgreSQL, Vercel AI SDK, Turborepo*

- Built an interactive **ER diagram canvas** (React Flow + Dagre auto-layout) rendering database and table nodes with column metadata and foreign-key edges, backed by a three-level PostgreSQL explorer with paginated table viewers and table sizes from system catalogs.
- Engineered a streaming **AI database assistant** on the Vercel AI SDK with 5 schema-aware tools and 2 model providers (Gemini, OpenRouter), enabling natural-language schema exploration and SQL/DDL drafting from live database metadata.
- Architected an **8-workspace Turborepo monorepo**, migrating the backend from NestJS to Hono with Zod-validated OpenAPI contracts and a typed RPC client, plus a shared LRU cache package with per-key TTL and in-flight promise coalescing.

ColorCraft Paint Visualizer | *React, TypeScript, HTML5 Canvas, Vite* Live Demo | GitHub

- Built a texture-preserving paint visualizer (**React, TypeScript, HTML5 Canvas**) that segments building regions from edge-detection images and applies paint blended with the original surface textures.
- Generated **normal-map-derived intensity maps** for realistic lighting on painted regions, with multi-region selection, paint-all, and preset/upload image pipelines.

Education

Master of Computer Applications Oct 2021 – Jun 2023
Francis Xavier's Engineering College *Tirunelveli, TN, India*

Bachelor of Computer Applications Jun 2018 – Apr 2021
St. John's College *Tirunelveli, TN, India*